EMT 1101 *Test # 2* November 15, 2013

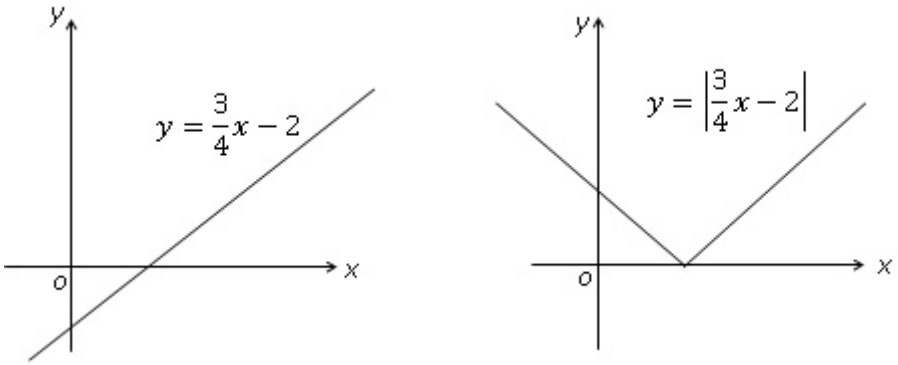
Answer the questions in the spaces provided on the question sheets. If you run out of room, continue on the last page. This test has **PAGES**

Time allowed : 2 hours

Name _____

Reg Number , Programme: _____

1. (a) The figure below shows the graph of $y = \frac{3}{4}x - 2$, and the graph of $y = |\frac{3}{4}x - 2|$.



i. Determine the x- and y- intercepts for each graph. (1)

Solution: For the first graph;
x-intercept is $0 = \frac{3}{4}x - 2$
$3x = 8$ from which $x = \frac{8}{3}$

$\frac{1}{2}$ mark

y-intercept is $y = 0 - 2 = -2$

$\frac{1}{2}$ mark

Second graph
x-intercept is $x = \frac{8}{3}$

$\frac{1}{2}$ mark

and y-intercept is $y = 2$

$\frac{1}{2}$ mark

ii. Which portions of the two graphs are identical? Give your answer in terms of an interval along the x-axis. (1)

Solution: The two graphs are identical in the interval $\left[\frac{8}{3}, \infty\right)$

iii. Explain how the graph of $y = |\frac{3}{4}x - 2|$ can be obtained from the graph of $y = \frac{3}{4}x - 2$ by means of reflection. (2)

Solution: $y = |\frac{3}{4}x - 2|$ can be obtained by reflecting the graph $y = \frac{3}{4}x - 2$, $\frac{8}{3} \leq x \leq \infty$ in the line $x = \frac{8}{3}$

(b) i. Find the sum and the product of the roots of the quadratic equation: $x^2 + 8x - 20 = 0$. (2)

Solution: If the roots are r_1, r_2 then $r_1 r_2 = -20$

1 mark

and $r_1 + r_2 = -8$

1 mark

ii. Find a quadratic equation with the given roots, $r_1 = 1-\sqrt{2}$ and $r_2 = 1+\sqrt{2}$. Write your answer in the form $ax^2+bx+c = 0$, where a, b, and c are integers and $a > 0$. (2)

Solution: If $r_1 = 1 - \sqrt{2}$ and $r_2 = 1 + \sqrt{2}$ then in $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
for solution to $ax^2 + bx + c = 0$
$-b = 1,\ 2a = 1$ and $a = \frac{1}{2}$
so $b^2 - 4ac = 2 = 1 - 4\frac{1}{2}c$
$-2c = 1$ and $c = -\frac{1}{2}$

or

$(x - 1 - \sqrt{2})(x - 1 + \sqrt{2}) = 0$
$(x-1)^2 - (\sqrt{2})^2 = 0$
$x^2 - 2x + 1 - 2 = 0$
$x^2 - 2x - 1 = 0$

iii. Find the value of k such that the equation $x^2 + 12x + k = 0$ has exactly one real root. (1)

Solution: If $x^2 + 12x + k = 0$ has exactly one root, then let $a^2 = k$ and so $x^2 + 12x + a^2 = 0$
$(x + a)^2 = 0$, $12x = 2ax$, therefore $a = 6$ and so $a^2 = 36$. Hence $k = 36$.

Alternatively
IF the equation has one root then $b^2 - 4ac = 0$, so $12^2 - 4k = 0$, i.e. $144 - 4k = 0$
So $4k = 144$ from which $k = 36$

2. Determine all real-number solutions for the equation $\sqrt{7 - 3x} - 6(x + 1) = 1$, then check for extraneous solutions. (10)

Solution: $\sqrt{7-3x}-6x-6=1 \Rightarrow \sqrt{7-3x}=6x+7 \Rightarrow 7-3x=36x^2+84x+49 \Rightarrow 36x^2+87x+42=0 \Rightarrow 12x^2+29x+14=0 \Rightarrow (2+3x)(7+4x)=0 \Rightarrow x=\frac{-2}{3}$, or $x=\frac{-7}{4}$
[5]

Checking for extraneous solutions:
If $x=\frac{-2}{3}$, then LHS$=\sqrt{7-3(-2/3)}-6(-2/3+1)=\sqrt{7+2}+4-6=\sqrt{9}-2=3-2=1=$RHS. Therefore, $x=\frac{-2}{3}$ is a solution to $\sqrt{7-3x}-6(x+1)=1$.**[2]**

If $x=\frac{-7}{4}$, then LHS$==\sqrt{7-3(-7/4)}-6(-7/4+1)=\sqrt{7+21/4}+\frac{21}{2}-6=\sqrt{49/4}+\frac{9}{2}=\frac{7}{2}+\frac{9}{2}=\frac{16}{2}=8\neq$RHS. Therefore, $x=\frac{-7}{4}$ is an extraneous solution.**[3]**

3. Solve for x and use interval notation to give your answer:

(a) $x^2 + x - 1 > 0$ (3)

Solution: $x^2 + x - 1 = 0$
$\Rightarrow x = \frac{-1 \pm \sqrt{1+4}}{2}$
$\Rightarrow$ the key numbers for $x^2 + x - 1 > 0$ are $\frac{-1-\sqrt{5}}{2}$ and $\frac{-1+\sqrt{5}}{2}$
Since the graph of the quadratic is concave up, $x^2 + x - 1 > 0, \forall x \in (-\infty, \frac{-1-\sqrt{5}}{2}) \cup (\frac{-1+\sqrt{5}}{2}, \infty)$

(b) $|\frac{x+1}{2} - \frac{x-1}{3}| < 1$ (3)

Solution: $|\frac{x+1}{2} - \frac{x-1}{3}| < 1$
$\Rightarrow |\frac{3x+3-2x+2}{6}| < 1$
$\Rightarrow |\frac{x+5}{6}| < 1$
$\Rightarrow \frac{x+5}{6} > -1$ and $\frac{x+5}{6} < 1$
$\Rightarrow x + 5 > -6$ and $x + 5 < 6$
$\Rightarrow x > -11$ and $x < 1$
$\Rightarrow x \in (-11, 1)$

(c) A piece of wire 12 cm long is cut into two pieces. Denote the lengths of the two pieces by x and $12 - x$. Both pieces are then bent into squares. For which values of x will the combined areas of the squares exceed 5 cm^2? (4)

Solution: $(\frac{x}{4})^2 + (\frac{12-x}{4})^2 > 5$
$\Rightarrow \frac{x^2}{16} + \frac{144-24x+x^2}{16} > 5$
$\Rightarrow 2x^2 - 24x + 144 > 80$
$\Rightarrow x^2 - 12x + 32 > 0$
$\Rightarrow (x - 4)(x - 8) > 0$
$\Rightarrow$ both factors must be negative or both factors must be positive
$\Rightarrow x < 4$ or $x > 8$
i.e. $x \in (0, 4) \cup (8, 12)$

4. (a) Find the domain of $f(x) = \dfrac{x^3+5}{\sqrt{x^2-x-2}}$ (6)

Solution: We want $x^2 - x - 2 > 0$, i.e. $(x-2)(x+1) > 0$
The key numbers are $x = -1$ and $x = 2$. So we obtain the intervals $(-\infty, -1), (-1, 2), (2, \infty)$ from which we have

	$(-\infty, -1)$	$(-1, 2)$	$(2, \infty)$
Test value	-2	0	3
$x+1$	-	+	+
$x-2$	-	-	+
$(x+1)(x-2)$	+	-	+

Therefore $x^2 - x - 2 > 0$ for all real numbers in $(-\infty, -1) \cup (2, \infty)$. Hence the domain of $f(x)$ is $(-\infty, -1) \cup (2, \infty)$

(b) The fixed point of a function $f(x)$ is a number x_0 such that $f(x_0) = x_0$. Find the fixed points of $f(x) = \sqrt{\frac{5}{4}x^2 - 9}$ (4)

Solution: $f(x_0) = x_0$ implies
$\sqrt{\frac{5}{4}x_0^2 - 9} = x_0$

$\frac{5}{4}x_0^2 - 9 = x_0^2$

$\frac{1}{4}x_0^2 - 9 = 0$

$\left(\frac{1}{2}x_0 - 3\right)\left(\frac{1}{2}x + 3\right) = 0$

$x_0 = 6$ or $x_0 = -6$

5. Let $f(x) = 3x + 1$ and $g(x) = x^2 - 4$. Compute the following.

(a) $f(g(x))$ (3)

Solution: $f(g(x)) = f(x^2 - 4) = 3(x^2 - 4) + 1 = 3x^2 - 11$

(b) $f(g(2))$ (2)

Solution: $f(g(2)) = 3(2)^2 - 11 = 1$

(c) $g(f(x))$ (3)

Solution: $g(f(x)) = g(3x + 1) = (3x + 1)^2 - 4 = 9x^2 + 6x - 3$

(d) $g(f(2))$ (2)

Solution: $g(f(2)) = 36 + 12 - 3 = 45$

6. Let

$$f(x) = |x|, \qquad F(x) = 1/x, \qquad g(x) = \sqrt{1 - x^2}.$$

Sketch the graph of each of the following functions.

(a) $y = f(5 - x)$ (3)

(b) $y = -F(x + 3)$ (3)

(c) $y = 1 - g(x - 2)$ (4)

HINT: This is best done by starting with the basic function. Give enough detail in your sketch to identify the position of the transformed curve.

Solution: